Linux Minimal FileSystem

FOUNDATIONS OF EMBEDDED SYSTEMS

בלינוקס Root File System

- The entire Linux directory structure starting at the top (/) root directory.
- A specific type of data storage format, such as EXT3, EXT4, BTRFS, XFS, and so on. Linux supports almost 100 types of filesystems, including some very old ones as well as some of the newest. Each of these filesystem types <u>uses its own metadata structures</u> to define <u>how the data is stored</u> and <u>accessed</u>.
- A <u>partition</u> or logical volume formatted with a <u>specific type of filesystem</u> that can be mounted on a specified <u>mount point on a Linux filesystem</u>

צורה סטנדרטית לארגון תיקיות ה- Root File System (1)

Directory	**Description**
/ (root filesystem)	The root filesystem is the top-level directory of the filesystem. It must contain all of the files required to boot the Linux system before other filesystems are mounted. It must include all of the required executables and libraries required to boot the remaining filesystems. After the system is booted, all other filesystems are mounted on standard, well-defined mount points as subdirectories of the root filesystem.
/bin	The /bin directory contains **user executable** files.
/boot	Contains the static bootloader and kernel executable and configuration **files required to boot a Linux computer**.
/dev	This directory contains the device files for every hardware device attached to the system. These are not device drivers, rather they are files that represent each device on the computer and facilitate access to those devices.
/etc	Contains the local system configuration files for the host computer.
/home	Home directory storage for user files. **Each user** has a subdirectory in /home.

צורה סטנדרטית לארגון תיקיות ה- Root File System (2)

/lib	Contains shared library files that are required **to boot the system**.
/media	**A place to mount external removable media devices** such as USB thumb drives that may be connected to the host.
/mnt	**A temporary mountpoint** for regular filesystems (as in not removable media) that can be used while the administrator is repairing or working on a filesystem.
/opt	Optional files such as vendor supplied application programs should be located here.
/root	This is not the root (/) filesystem. It is the **home directory for the root user**.
/sbin	System binary files. These **are executables used for system administration**.
/tmp	**Temporary directory**. Used by the operating system and many programs to store temporary files. Users may also store files here temporarily. Note that files stored here may be deleted at any time without prior notice.
/usr	These are shareable, read-only files, including executable binaries and libraries, man files, and other types of documentation.
/var	Variable data files are stored here. This can include things like log files, MySQL, and other database files, web server data files, email inboxes, and much more.

צורה סטנדרטית לארגון תיקיות ה- Root File System (3)

- pseudo filesystems
- מייצגות מידע של ה-kernel כקבצים בהיררכיה של תיקיות.
- כשקוראים אחד מהקבצים, המידע שרואים אינו מגיע מדיסק האחסון. המידע נוצר on the fly ע"י פונקציה בקרנל.
- חלק מהקבצים ניתנים לכתיבה, כלומר פונקציה בקרנל תיקרא עם המידע החדש שאותו אנו כותבים לקובץ ובמידה ויש הרשאות והמידע בפורמט תקין ישתנה הערך של השדה המתאים במרחב זיכרון הקרנל.
- proc – חשיפת מידע על כל התהליכים ל- user space, הצגת מידע ויכולת שליטה על מצב והתנהגות תתי מערכות בקרנל.
- sysfs – הצגת הירכית הדרייברים וההתקנים ל- user space. הצורה בה ההתקנים מחוברים אחד לשני.

THE ABSOLUTE MINIMUM

THE FUNDAMENTAL CHALLENGE

What is the absolute minimum technical component required for a Linux Root File System?

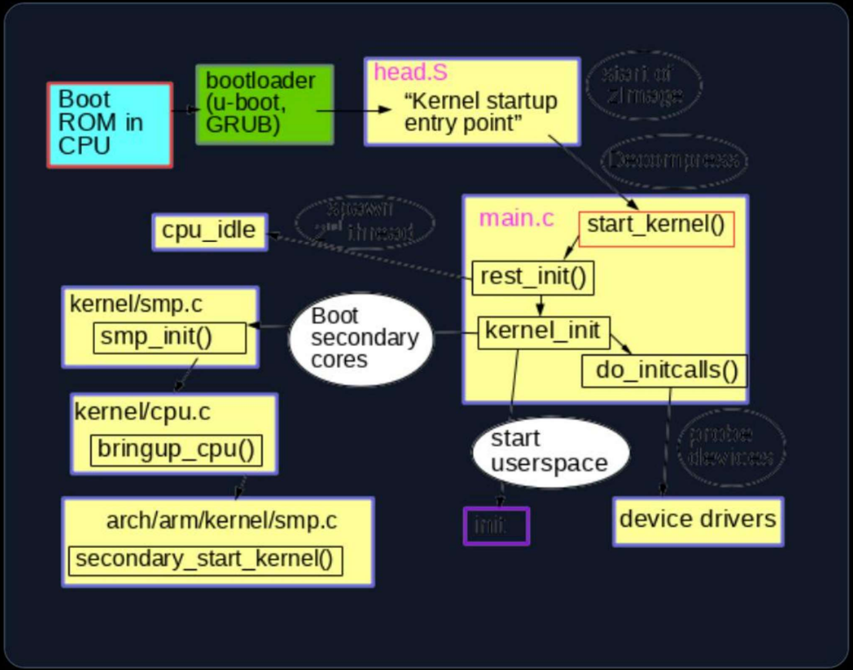
VERIFIED IMPLEMENTATION

```
int main()
{
    while(1);
    return 0;
}
```

The Prerequisite:

At its core, the Kernel only requires a single executable file (Init) that prevents a Kernel Panic by implementing an infinite operational loop.

SYSTEM HIERARCHY



The Boot Handshake

01. Kernel Init: Kernel initializes hardware and mounts the RootFS.

02. PID 1: Kernel looks for the binary at `/sbin/init`.

03. UserSpace: The first executable takes control of system execution.

PRACTICAL LAB ASSIGNMENT

Absolute Minimum RootFS Construction

A guided workshop on implementing an absolute minimum rootfs

`[root@minimal-fs ~]# _`

CORE COMPONENTS

BusyBox Utility

Reduces footprint by combining multiple Unix utilities (ls, cat, grep) into one multi-call binary.

C Library

Essential for standard functions. Minimalist choices include musl or uClibc-ng for reduced footprint.

Command Shell

Typically ash or sh, providing a basic interface for system interaction and scripting.

Static Linking

Compiling binaries with integrated libraries to eliminate the need for `/lib` nodes.

PRACTICAL HARDENED SYSTEMS

Bespoke Init Architecture: Create a custom Init application that uses `fork()` and `execv()` for every intended application, ensuring strict process control.

Attack Surface Reduction: Do **not** mount `/sys`, `/proc`, or `/dev`. This prevents attackers from performing system reconnaissance.

Secure IPC: Use Linux **pipes** for inter-process communication instead of shared files or network sockets.

Filesystem Hygiene: Eliminate all unnecessary files. Use **static linking** (uClibc) for all binaries to keep the filesystem truly minimal.

HARDENED PATTERN

```
// Example: Isolated Launcher
pid_t pid = fork();
if (pid == 0) {
    char *args[] = {"/app", NULL};
    execv(args[0], args);
    _exit(1);
}
// Communication via pipes follows...
```

Questions ?

Ready for hardened implementation.

```
[root@minimal-fs ~]# _
```